MILITARY SPECIFICATION

RELAYS, TELEGRAPH, PASSIVE, SOLID-STATE, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for completely solid-state passive relays for use in telegraph circuits and associated equipment.

1.2 Classification.

1.2.1 Types. Relays shall be of the following types (see 6.2):

Type I - Nonrepairable, sealed (see 3.4.1.1.1).
Type II - Nonrepairable, unsealed encapsulated (see 3.4.1.1.2).

1.2.2 Grades. Relays shall be of the following grades (see 6.2):

Grade A - Qualified relays
Grade B - Qualified relays with a 40,000 hour mean-time-between-failure (MTBF). This failure rate level is established at a confidence level of 90 percent for qualification and 10 percent for maintenance of qualification

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Steel, Flat and Seals.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Fiberboard.
PPP-B-676 - Boxes, Setup.
PPP-T-60 - Tape: Packaging, Waterproof.
PPP-T-76 - Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing).
NN-P-71 - Pallet, Material Handling, Wood, Double Faced, Stringer Construction.

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MIL-M-14	-	Molding Plastics and Molded Plastic Parts, Thermosetting.
MIL-P-116	-	Preservation, Methods of.
MIL-P-997	-	Plastic Material, Laminated, Thermosetting, Electrical Insulation: Sheets, Glass Cloth, Silicone Resin.
MIL-P-15037	-	Plastic Sheet, Laminated, Thermosetting, Glass-Cloth, Melamine - Resin.
MIL-P-15047	-	Plastic-Material, Laminated Thermosetting Sheets, Nylon Fabric Base, Phenolic - Resin.
MIL-B-43014	-	Boxes: Water Resistant Paperboard; Folding, Set-Up, and Metal-Stayed.
MIL-C-45662	-	Calibration System Requirements.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-147	-	Palletized Unit Loads on 40" x 48" Pallets.
MIL-STD-188	-	Military Communication System Technical Standards.
MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454	-	Standard General Requirements for Electronic Equipment.
MIL-STD-461	-	Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-1285	-	Marking of Electrical and Electronic Parts.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ELECTRONIC INDUSTRIES ASSOCIATION

RS-209-A - EIA Standards for Electron Tubes.

(Application for copies should be addressed to Electronic Industries Association, Engineering Department, Standard Sales, 2001 Eye Street, N.W., Washington, D.C. 20006.)

AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI Y32.2 - Graphics Symbols for Electrical and Electronics Diagrams.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway New York, N.Y. 10018.)

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of conflict between requirements of this specification and the specification sheets, the latter shall govern (see 6.2).

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3.2 Qualification. Relays furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for the opening of the bids (see 4.5 and 6.3). If relays are qualified with a 40,000 hour MTBF, the grade B life test shall be as specified in table IV.

3.3 Materials. The materials shall be as specified herein. However, when materials are other than specified herein, materials shall be used which will enable the relays to meet the requirements of this specification (see 4.3).

3.3.1 Fungus-resistance. External materials used in the construction of relays shall be fungus inert (see requirement 4 of MIL-STD-454).

3.3.2 Plastic. Laminated plastic material shall conform to MIL-P-997, MIL-P-15037, or MIL-P-15047. Molded plastic material shall conform to MIL-M-14. Cotton-filled or wood-flour-filled materials shall not be used.

3.3.3 Metals. Metals shall be of a corrosion-resistant type or shall be plated or treated to resist corrosion.

3.3.3.1 Dissimilar metals and compatible couples. When dissimilar metals are used in intimate contact with each other, protection against electrolysis and corrosion shall be provided. The use of dissimilar metals in contact, which tend toward active electrolytic corrosion (particularly brass, copper, or steel used in contact with aluminum or aluminum alloy) is not acceptable. However, metal-plating or metal spraying of dissimilar-base metals to provide similar or suitable abutting surfaces if permitted. The use of dissimilar metals separated by a suitable insulating material is also permitted. Dissimilar metals and compatible couples are defined in 6.4 through 6.4.4 and table IX.

3.4 Design and construction. Relays shall be of the design, construction, and physical dimensions specified (see 3.1).

3.4.1 Modes of operation. Relays (see 1.2.1) shall be designed and constructed to provide for operation of at least one of the following modes (see 3.1):

(a) Neutral input - Neutral output:
    (1) One each receiving module (see 6.5.6).
    (2) One each transmitting module (see 6.5.7).
(b) Polar input - Polar/neutral output:
    (1) Two each receiving modules.
    (2) Two each transmitting modules.
    (3) One each bias module (see 6.5.5).
(c) Neutral/biased input - Polar/neutral output:
    (1) Two each receiving modules.
    (2) Two each transmitting modules.
    (3) One each bias module.

3.4.1.1 Types. These relays need not necessarily be constructed using modular design but must provide for operation in the modes specified in 3.4.1. Relays shall be of the following types (see 6.2):

3.4.1.1.1 Type I. Nonrepairable sealed unit.

3.4.1.1.2 Type II. Nonrepairable encapsulated unsealed unit. The base of the relay shall be a separate subassembly, detached from the unit within the case. The electrical wiring shall be brought through the bottom of the unit to enable wiring in any of the various modes. Initial wiring to the base shall be as specified by the procuring activity (see 6.2).

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3.4.2 Wiring. On all units, the wiring to the base shall be as specified (see 3.1), and identified by color symbol as shown in table I.

3.4.3 Case. The case shall be constructed of metal that will provide electrostatic and electromagnetic shielding. Unless otherwise specified, the case shall be connected to pin 5 to provide means for grounding (see 3.1).

3.4.4 Contact polarity. The output contacts shall be nonpolarized.

3.4.5 Operating power. The relay operating power shall be derived only from the incoming signal line, and bias line where bias current is required.

3.4.6 Weight. The weight shall be as specified (see 3.1).

3.4.7 System compatibility. Unless otherwise specified, relays shall conform to the requirements of MIL-STD-188 with respect to waveshape, input sensitivity (6.5.12), neutral operation (6.5.9), and polar relays (6.5.10).

TABLE I. Color symbols.

Color	Significant figure	Color abbreviation
Black	0	Blk
Brown	1	Brn
Red	2	Red
Orange	3	Orn
Yellow	4	Yel
Green	5	Grn
Blue	6	Blu
Violet	7	Vio
Gray	8	Gy
White	9	Wht

3.5 Electrical characteristics. When relays are tested as specified in 4.7.2, the electrical characteristics shall be as specified; typical characteristics are shown in table II.

TABLE II. Electrical and technical characteristics (at 25°C).

Characteristics	Typical limit
Input current:	
Polar	15 to 80 mA
Neutral	10 to 80 mA
Neutral/biased	20 to 80 mA (signal); 10 to 40 mA (biased) Input 1 must be electrically symmetrical with input 2
Input resistance	Less than 250 ohms at 60 mA
Keying rate	Continuous dc to 1200 baud
Bias (pulse-width) distortion	Less than 1 percent at 400 baud, distortion adjustable with bias current in "neutral/biased" mode
Output:	
Current	20 to 100 mA neutral/polar
Voltage	260 V, continuous in neutral mode ±130 V; continuous in polar or neutral/biased modes, 400 V for 200 milliseconds
Contact	Form K nonpolarized (null seeking with no bias or signal input)
Open-contact leakage current	Shall not exceed 10 μA at 260 V, 25°C
Closed-contact voltage drop	Less than 3 volts
Operating power	Loop current only for neutral and polar inputs; loop current and bias current for the "neutral/biased" mode
Leakage-current	Less than 10 microamperes (μA) at 260 V
Applications	See 3.1
Capacitance (when specified)	See 3.1

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3.5.1 Input resistance. When relays are tested as specified in 4.7.2.1, the input resistance shall be as specified (see 3.1). Where polar or differential operation is used, the two input sections shall be symmetrical and interchangeable.

3.5.2 Open-contact leakage current. When relays are tested as specified in 4.7.2.2, the open-contact leakage current shall not exceed 10 μA.

3.5.3 Closed-contact voltage drop and stability. When relays are tested as specified in 4.7.2.3, contact voltage drop shall not exceed 3 volts, and the contacts shall not deviate from the actuated state when the contact current is varied or removed and replaced. When a steady "mark" (see 6.5.3) current is applied to the input, the contacts shall remain closed.

3.5.4 Keying rate. When relays are tested as specified in 4.7.2.4, they shall operate satisfactorily at speeds up to 1,200 bauds.

3.5.5 Bias (pulse-width) distortion (see 6.5.8). When tested as specified in 4.7.2.4, pulse-width distortion shall not exceed 1 percent at 400 bauds or 3 percent at 1,200 bauds (see 6.5.1).

3.5.6 Output resistance and sensitivity. When relays are tested as specified in 4.7.2.5, the specified input currents (see 3.1) shall cause the output to transfer to the operating condition and to revert to the nonoperating condition.

3.5.7 Capacitance (when specified, see 3.1). When relays are tested as specified in 4.7.2.6, the capacitance shall not exceed the value specified.

3.6 Thermal shock. When relays are tested as specified in 4.7.3, there shall be no evidence of mechanical damage. Electrical characteristics shall be measured following the test and shall remain as specified.

3.7 High and low temperature. When relays are tested as specified in 4.7.4, they shall operate without failure.

3.8 Moisture resistance. When relays are tested as specified in 4.7.5, there shall be no evidence of deterioration which would adversely affect subsequent applications of the relay, as determined by visual inspection, insulation resistance, dielectric withstanding voltage, electrical characteristics tests and marking shall be legible. Insulation resistance shall meet the requirements of 3.13.

3.9 Vibration, high frequency. When relays are tested as specified in 4.7.6, there shall be no misses during monitoring, the relays shall meet the requirements of bias distortion, and there shall be no evidence of loosening of parts.

3.10 Shock (specified pulse). When relays are tested as specified in 4.7.7, they shall meet requirements of bias distortion and there shall be no evidence of loosening of parts or other damage.

3.11 Radio frequency interference test. When relays are tested in accordance with 4.7.8, they shall meet the requirements of MIL-STD-461 for class 1C equipment and shall be as specified herein. All tests shall be performed with the relay operating in the test circuit shown in figure 1.

3.12 Dielectric withstanding voltage. When relays are tested as specified in 4.7.9, they shall withstand the application of 500 volts rms, 60 Hz without damage or breakdown; leakage current shall not exceed 100 microamperes.

3.13 Insulation resistance and isolation. When relays are tested as specified in 4.7.10, the insulation resistance between all insulated terminals and case or frame shall be 500 megohms or greater, and the isolation between circuits shall be as follows:

(a) All input circuit terminals connected together and all output circuit terminals connected together - 50 megohms or greater.
(b) Terminals of one input circuit connected together and terminals of the other input circuit connected together - 50 megohms or greater.

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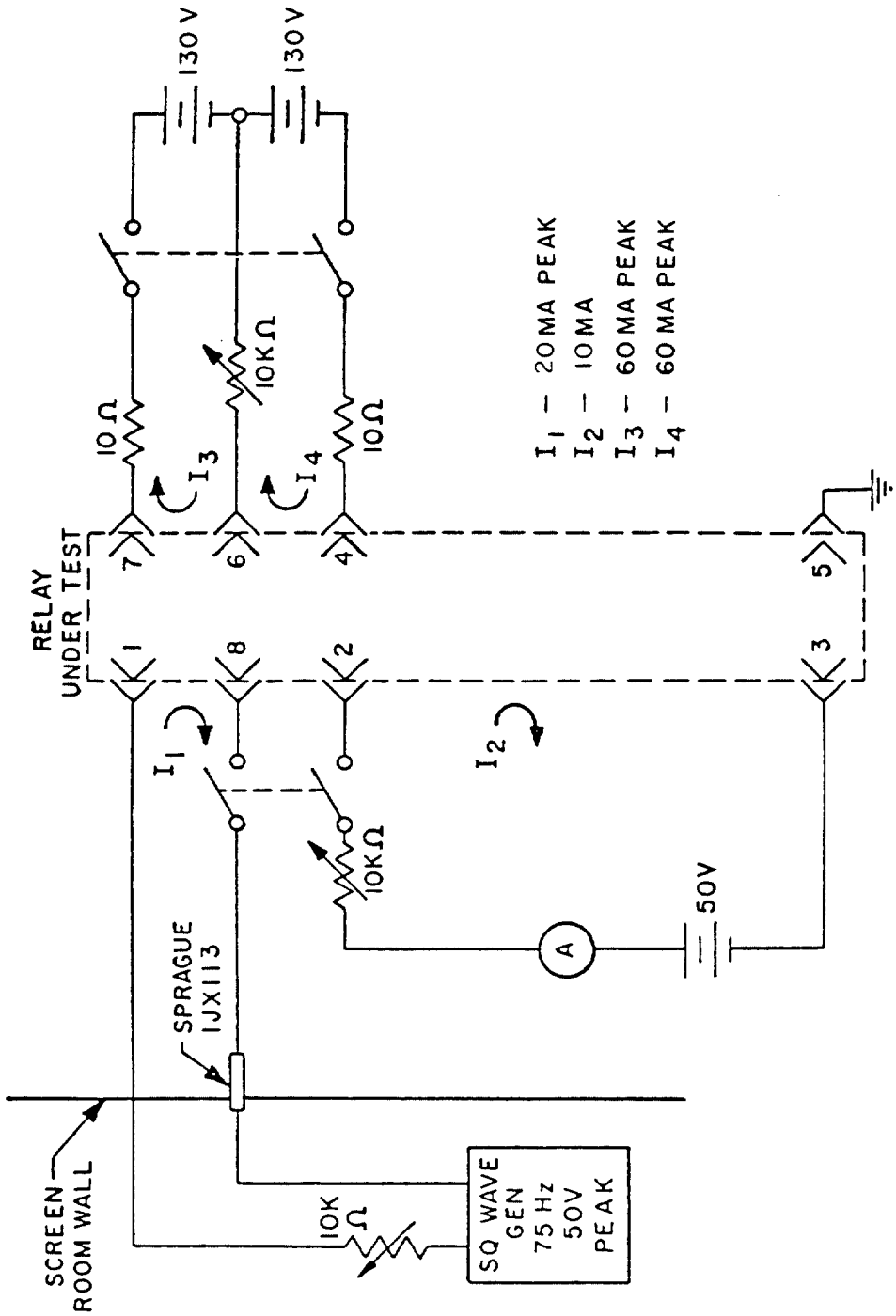


FIGURE 1. Test circuit.

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3.14 Terminal strength. When relays are tested as specified in 4.7.11, the terminals shall not loosen nor shall there be any other damage. Bending of the terminals shall not be construed as damage provided they can be reformed in a manner to permit proper mating with the socket, and the sealing, if applicable, is not affected by the process.

3.15 Seal (when applicable) (see 3.1). When relays are tested as specified in 4.7.12, the maximum total leakage of gas atmosphere shall not exceed a rate of 1×10^{-6} atmospheric cubic centimeters per second (atm cc/sec) per cubic inch of net sealed volume, for relays over 2 cubic inches in volume. For relays of 2 cubic inches net sealed volume or less, the leakage rate shall not exceed 1×10^{-8} (atm cc/sec). When specified (see 3.1), relays shall be hermetically sealed (see 6.5.2).

3.16 Altitude. When the relays are tested as specified in 4.7.13, they shall operate without failure.

3.17 Life. When relays are tested as specified in 4.7.14, there shall be no misses during monitoring periods and relays shall meet the requirements of dielectric withstanding voltage, insulation resistance, and the electrical tests as specified in 3.12, 3.13, and 3.5 through 3.5.7.

3.18 Marking (see 3.1). The following information shall be marked on the relays in accordance with MIL-STD-1285. Symbols shall be in accordance with ANSI Y32.2. As a minimum, the following information shall be included:

(a) Part number (see 3.1).
(b) Circuit diagram showing contacts and input-output symbolism with necessary terminal identification.
(c) Manufacturer's name or code symbol and date code.
(d) Suffix shall be added to the part number for relays that have been subjected to and meet life test (see 3.17).

3.19 Soldering. When soldering is employed, the use of flux shall be avoided whenever possible. If the use of flux is warranted, only noncorrosive fluxes shall be used unless it can be shown that nondestructive methods can eliminate all traces of corrosive elements after soldering. Soldering shall be in accordance with requirement 5 of MIL-STD-454.

3.20 Workmanship. The relays shall be fabricated in such a manner as to be uniform in quality, and shall be free from cracked or displaced parts, sharp edges, burrs, and other defects that will affect life, serviceability, and appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to permit performance of the required inspection shall be established and maintained by the supplier. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-C-45662.

4.2 Classification of inspection. The inspections specified herein are classified as follows:

(a) Materials inspection (see 4.3).
(b) Qualification inspection (see 4.5).
(c) Quality conformance inspection (see 4.6).

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4.3 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials listed in table III, used in fabricating the relays, are in accordance with the applicable referenced documents prior to such fabrication.

TABLE III. Materials inspection.

Materials	Requirement paragraph	Applicable documents
Fungus-inert materials Plastic	3.3.1 3.3.2	MIL-STD-454 MIL-P-997, MIL-P-15037 MIL-P-15047, MIL-M-14

4.4 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202.

4.4.1 Direct current voltages. When a direct current potential is required for performing a given test and the test can be performed with the test potential connected in either polarity, the polarity used will be optional, unless both polarities are specified for the test.

TABLE IV. Qualification inspection.

Examination or test	Requirement paragraph	Method paragraph	Sample unit											
			1	2	3	4	5	6	7	8	9	10	11	12
Visual and mechanical inspection	3.1, 3.3, 3.4, 3.18, and 3.20	4.7.1	X	X	X	X	X	X	X	X	X	X	X	X
Electrical characteristics	3.5	4.7.2	X	X	X	X	X	X	X	X	X	X	X	X
Thermal shock	3.6	4.7.3	X	X	X	X					X	X	X	X
High and low temperature	3.7	4.7.4					X	X	X	X				
Moisture resistance	3.8	4.7.5	X	X	X	X								
Vibration, high frequency	3.9	4.7.6	X	X	X	X								
Shock (specified pulse)	3.10	4.7.7									X	X	X	X
Radio frequency interference control	3.11	4.7.8						X	X					
Dielectric withstanding voltage	3.12	4.7.9	X	X	X	X	X	X	X	X	X	X	X	X
Insulation resistance and isolation	3.13	4.7.10	X	X	X	X	X	X	X	X	X	X	X	X
Terminal strength	3.14	4.7.11					X			X			X	X
Seal (when applicable)	3.15	4.7.12	X	X	X	X	X	X	X	X	X	X	X	X
Visual and mechanical inspection	3.1, 3.3, 3.4, 3.18, and 3.20	4.7.1	X	X	X	X	X	X	X	X	X	X	X	X
Altitude	3.16	4.7.13	X								X			
Life 1/	3.17	4.7.14	X	X	X	X	X	X	X	X	X	X	X	X
Electrical characteristics	3.5	4.7.2	X	X	X	X	X	X	X	X	X	X	X	X

1/ Grade B. The sample units shall be combined with additional units to meet the following life test hours. No relay will be tested to more than 4,440 hours.

Number of failures permitted	Cumulative hours
0	92,125
1	155,625
2	212,875
3	267,250
4	319,750
5	370,937

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4.5 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.3) on sample units produced with equipment and procedures normally used in production.

4.5.1 Sample size. The number of relays to be subjected to qualification inspection shall be as specified in table IV. For grade B life test only, the additional sample units required shall be as specified (see table IV).

4.5.2 Inspection routine. Sample units shall be subjected to the qualification inspection specified in table IV, in the order shown.

4.5.2.1 Post-tests. A test procedure (e.g., dielectric withstanding voltage) required at the conclusion of a procedure shall be known as a post-test of that procedure. Some post-tests are required to be repeated on sample units as post-tests of more than one environmental test procedure. Such a post-test shall be required only at the conclusion of the last environmental test procedure requiring it.

4.5.3 Failures. One or more failures shall be cause for refusal to grant qualification approval, except for grade B life test the test failures shall meet the requirements of table IV. Grade B relays for life test only shall demonstrate a maximum rate of 2.5 percent per 1,000 hours operation at a 90 percent confidence level.

4.5.4 Test samples and test data. Grade A and B relays tested to table IV together with certified test reports and test data shall be forwarded to the activity responsible for qualification (see 6.3). Grade B relays that have been subject to and passed life test only (see table IV) may be delivered on a contract or purchase order (see 3.18).

4.5.5 Retention of qualification. To retain qualification, the supplier shall forward a report at 36-months intervals for group A and 6-months intervals for group B to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall consist of:

(a) A summary of the results of the tests performed for inspection of product for delivery group A, indicating as a minimum the number of lots that have passed and the number that have failed. The results of tests of all reworked lots shall be identified and accounted for.
(b) A summary of the results of the tests performed for qualification verification inspection groups B and C, including the number and mode of failures. The summary shall include results of all qualification verification inspection tests performed and completed during the 36-month period for group A and the 6-month period for group B. If the summary of the test results indicates nonconformance with specification requirements, and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list.

Failure to submit the report within 30 days after the end of each 6-month or 36-month period may result in loss of qualification for the product. In addition to the periodic submission of inspection data, the supplier shall immediately notify the qualifying activity at any time during the 6-month or 36-month period that the inspection data indicates failure of the qualified product to meet the requirements of this specification.

In the event that no production occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during two consecutive reporting periods there has been no production, the manufacturer may be required, at the discretion of the qualifying activity, to submit the relays to testing in accordance with the qualification inspection requirements.

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4.6 Quality conformance inspection.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A inspection.

4.6.1.1 Inspection lot. An inspection lot shall consist of all relays of the same design and materials, produced under essentially the same conditions, and offered for inspection at one time. The lot size shall not exceed 500 relays.

4.6.1.2 Group A inspection. Group A inspection shall consist of the examinations and test specified in table V, in the order shown.

4.6.1.2.1 Sampling plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level II. The AQL shall be 1.0.

4.6.1.2.2 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

TABLE V. Group A inspection.

Examination or test	Requirement paragraph	Method paragraph
Visual and mechanical 1/	3.1, 3.3, 3.4, 3.18 and 3.20	4.7.1
Dielectric withstanding voltage	3.12	4.7.9
Insulation resistance and isolation	3.13	4.7.10
Electrical characteristics	3.5	4.7.2

1/ Physical dimensions shall be measured on two sample units per lot.

4.6.2 Qualification verification inspection. Qualification verification inspection shall consist of groups B and C. Except where the results of these inspections show noncompliance with the applicable requirements (see 4.6.2.2.4), delivery of products which have passed group A shall not be delayed pending the results of these qualification verification inspections.

4.6.2.1 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table VI in the order shown, and shall be made on sample units from lots which have passed the group A inspection.

TABLE VI. Group B inspection.

Test	Requirement paragraph	Method paragraph
Thermal shock	3.6	4.7.3
High and low temperature	3.7	4.7.4
Moisture resistance	3.8	4.7.5
Vibration, high frequency	3.9	4.7.6
Shock (specified pulse)	3.10	4.7.7
Dielectric withstanding voltage	3.12	4.7.9
Insulation resistance and isolation	3.13	4.7.10
Terminal strength	3.14	4.7.11
Seal (when applicable)	3.15	4.7.12
Altitude	3.16	4.7.13

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4.6.2.1.1 Sampling plan. Four sample units shall be selected from each 12-months production for grade A, 6-months production for grade B, of all relays covered by the same part number. No failures shall be allowed.

4.6.2.1.2 Disposition of sample units. Sample units which have been subjected to group B inspections shall not be delivered on the contract or purchase order.

4.6.2.2 Group C inspection. Group C inspection shall consist of the test specified in table VII and shall be made on sample units which have passed group A inspection.

TABLE VII. Group C inspection.

Test	Requirement paragraph	Method paragraph	Grade A	Grade B	
			Failures permitted	Cumulative hours	No. of failures permitted
Life	3.17	4.7.14	0	4,218 21,281 44,000 69,812 97,312 126,062	0 1 2 3 4 5

4.6.2.2.1 Sampling plan.

(a) Grade A. Every 36 months, six sample units shall be subjected to life test.
(b) Grade B. Every 6 months, test samples shall be accumulated and placed on life test. The test sample size shall be determined by the manufacturer so that the unit hours generated meet the requirements specified in table VII.

4.6.2.2.2 Failures.

(a) Grade A. If one or more sample units fail to pass group C inspection, the sample shall be considered to have failed.
(b) Grade B. The sample units shall meet the requirements specified in table VII. No relay shall be on life test for more than 4,400 hours.

4.6.2.2.3 Disposition of sample units. Sample units which have been subjected to and passed group C inspection may be delivered on the contract or purchase order (see 3.18).

4.6.2.2.4 Noncompliance. If a sample fails to pass group B or C inspection, the supplier shall take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the Government, has been taken. After the corrective action has been taken, group B and C inspections shall be repeated on additional sample units (all inspections, or the inspection which the original sample failed, at the option of the Government). Group A inspection may be reinstituted; however, final acceptance shall be withheld until group B and C inspections have shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure and corrective action taken shall be furnished to the cognizant inspection activity and the qualifying activity.

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4.6.3 Inspection of proparation for delivery. The sampling and inspection of the preservation-packaging and interior package marking shall be in accordance with the groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.7 Methods of examination and test.

4.7.1 Visual and mechanical inspection. Relays shall be examined to verify that construction details such as physical dimensions, materials, marking, finish, and workmanship are in accor - dance with specified requirements. When this inspection is performed as a part of group A inspec-tion, the physical dimensions shall be checked on only two relays which shall be selected at random from the sample.

4.7.2 Electrical characteristics (see 3.5). The electrical characteristics shall be determined as specified herein.

4.7.2.1 Input resistance (see 3.5.1). The test circuit shall be as shown in figure 2 or equivalent. Measure the voltage across the input terminals at an input current of 60 mA and calculate the re-sistance (output loading is optional).

4.7.2.2 Open-contact leakage current (see 3.5.2). Connect the relay as shown in figure 2, or equivalent. Measure leakage current with 260 ± 10 Vdc applied. Repeat with the output polarity reversed.

4.7.2.3 Closed contact-voltage drop and stability (see 3.5.3). Connect the relay as shown in figure 2 or equivalent. Measure the closed contact voltage drop at an output current of 60 mA with an input current of 20 mA, both before and following a complete interruption of output current for 1 second minimum. Both outputs of universal relays shall be tested. Neutral relays (see 6.5.9) strapped for 60 mA operation shall be tested with input current of 40 mA.

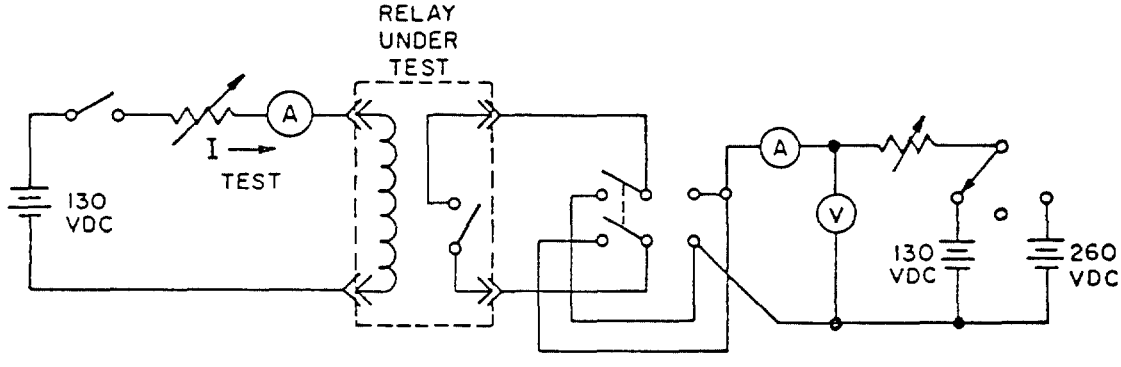


FIGURE 2. Test circuit for output voltage drop and input resistance and sensitivity.

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4.7.2.4 Bias (pulse-width) distortion (see 3.5.4 and 3.5.5). The test circuits shall be as shown in figures 3 through 6. The relays shall be tested at speeds of 400 ±2 percent bauds and 1,200 ±2 percent bauds with allowable pulse-width distortion of 1 percent and 3 percent, respectively. The relays shall be tested in accordance with the conditions in table VIII. Loop potentials shall be 130 volts dc for input signals. Output loops potentials shall be 260 volts dc for neutral output loops and 130 volts dc for polar loops. The input and output loop circuits shall be resistive (see figure 7). The input signal shall be a repeated step function with a 50-percent duty cycle. The output shall be monitored across a resistive test load. Using a dual trace oscilloscope, the input and output current waveforms shall be monitored to determine relative pulse-width. Neutral signal pulse-width measurements shall be made at the 50-percent amplitude points. Polar signal-widths shall be measured at the zero amplitude points. On universal relays repeat test with bias and input circuits transposed. Distortion shall be calculated as follows:

$$\text{Percent distortion} = \frac{[T_S \text{ ON} - T_S \text{ OFF}] \times 100}{T_W}$$

T_S ON is positive if the sampling point of the leading edge of the input pulse occurs in time prior to the sampling point of the leading edge of the device output pulse.

T_S OFF is positive if the sampling point of the trailing edge of the input pulse occurs in time prior to the sampling point of the trailing edge of the device output pulse.

Neutral relays strapped for 60 mA operations shall be tested under condition 3 only (see table VIII).

TABLE VIII. Bias distortion loading condition.

Condition	Input	Output
1	Minimum specified value	Maximum specified value
2	Mid specified value	Mid specified value
3	Maximum specified value	Minimum specified value

NOTE: Care must be taken to observe the additive or subtractive effect of the switch-times (see 6.5.14). The low allowable distortion with respect to the input pulse width prohibits viewing both switch times simultaneously. Therefore, when observing each transition, the test operator must determine the output pulse width plus the switch-times.

4.7.2.5 Output resistance and sensitivity (see 3.5.6). The test circuit shall be as shown in figure 2 or equivalent. With the input current equal to that specified (see 3.1), the output current shall be adjusted to 60 mA. The input current shall then be continuously reduced to 10 mA and the output voltage drop measured to verify that the output circuit is in the operate conditions. The input current shall then be reduced to 0 mA and the output shall be monitored to verify that the output circuit has transferred to the nonoperate condition. Universal relays (see 6.5.11) shall also be tested for operation in the neutral-biased mode using each input in turn as the bias input at 15 mA and at 30 mA bias current. A signal current variation of ±6 mA maximum about the bias current level shall be applied, and the relay shall be monitored for transfer between operating and nonoperating conditions. Neutral relays strapped for 60 mA operation will be tested only at input currents of 60 mA to 40 mA.

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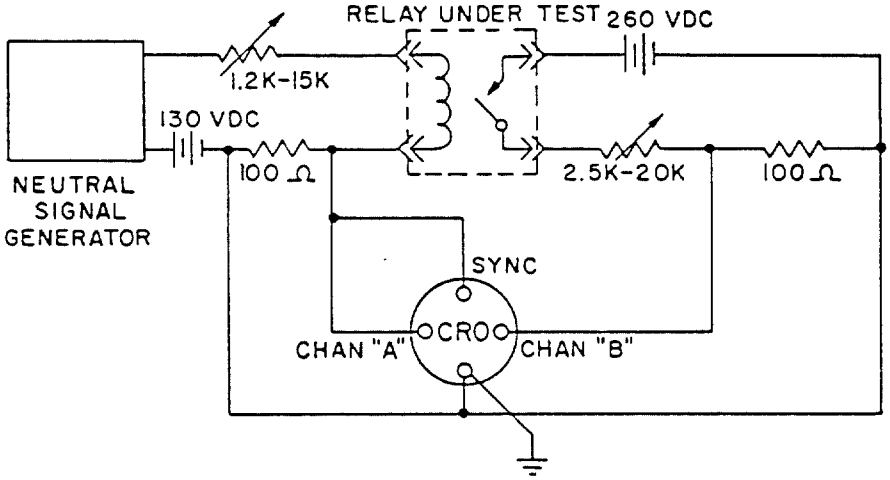


FIGURE 3. Distortion test circuit - neutral/neutral operation.

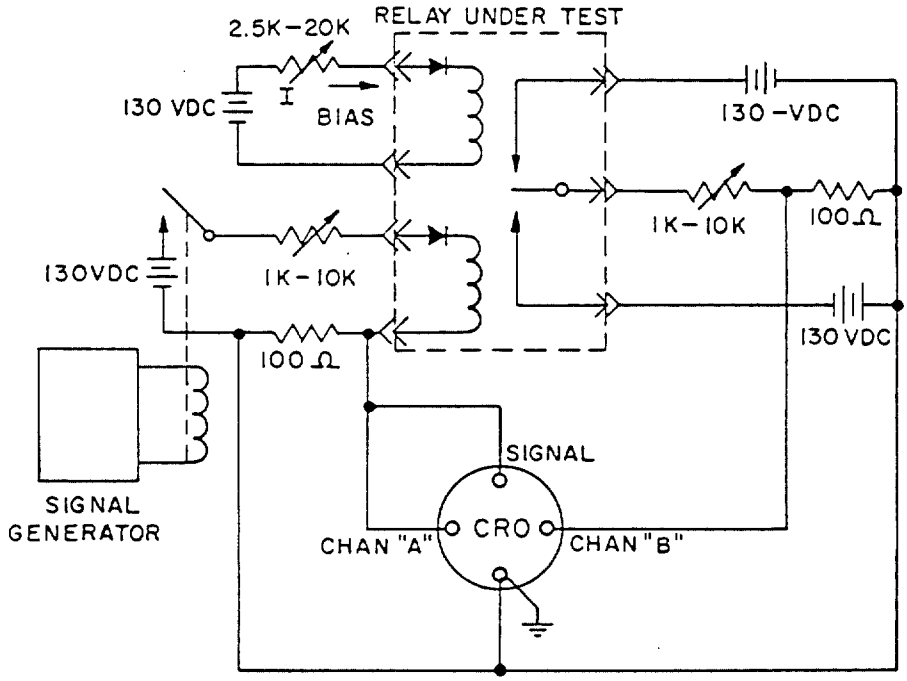


FIGURE 4. Distortion test circuit - neutral-bias/polar or neutral.

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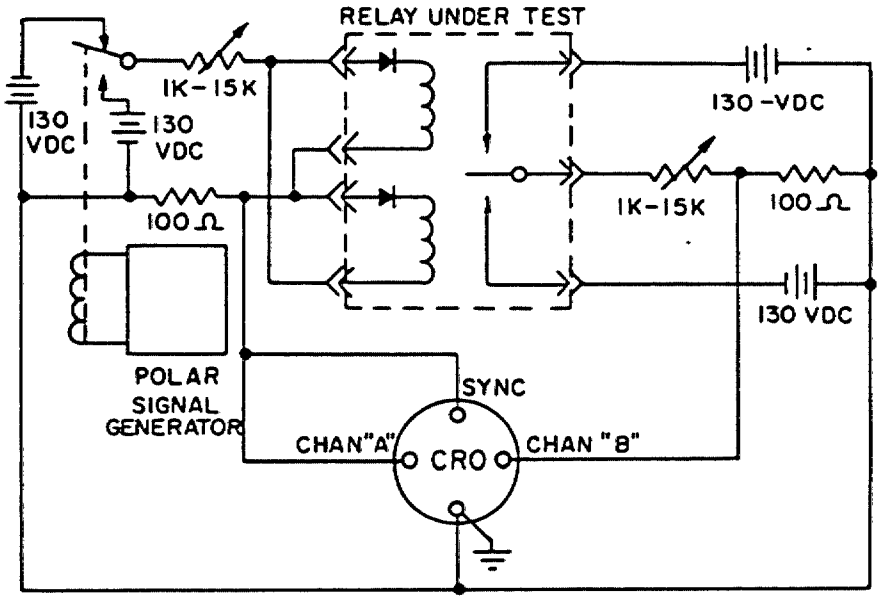


FIGURE 5. Distortion test circuit - 2-wire polar/polar or neutral.

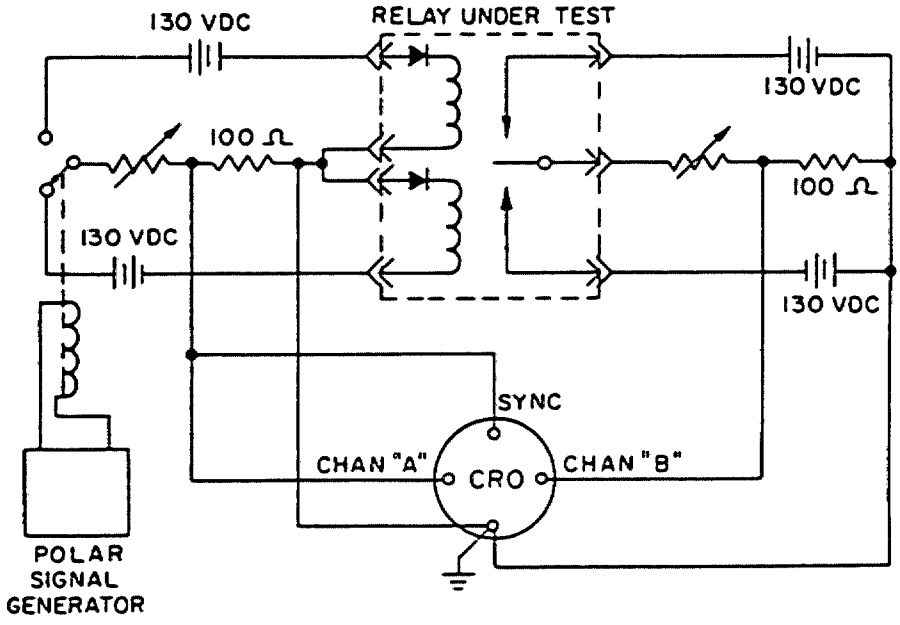


FIGURE 6. Distortion test circuit - 3-wire polar/neutral or polar.

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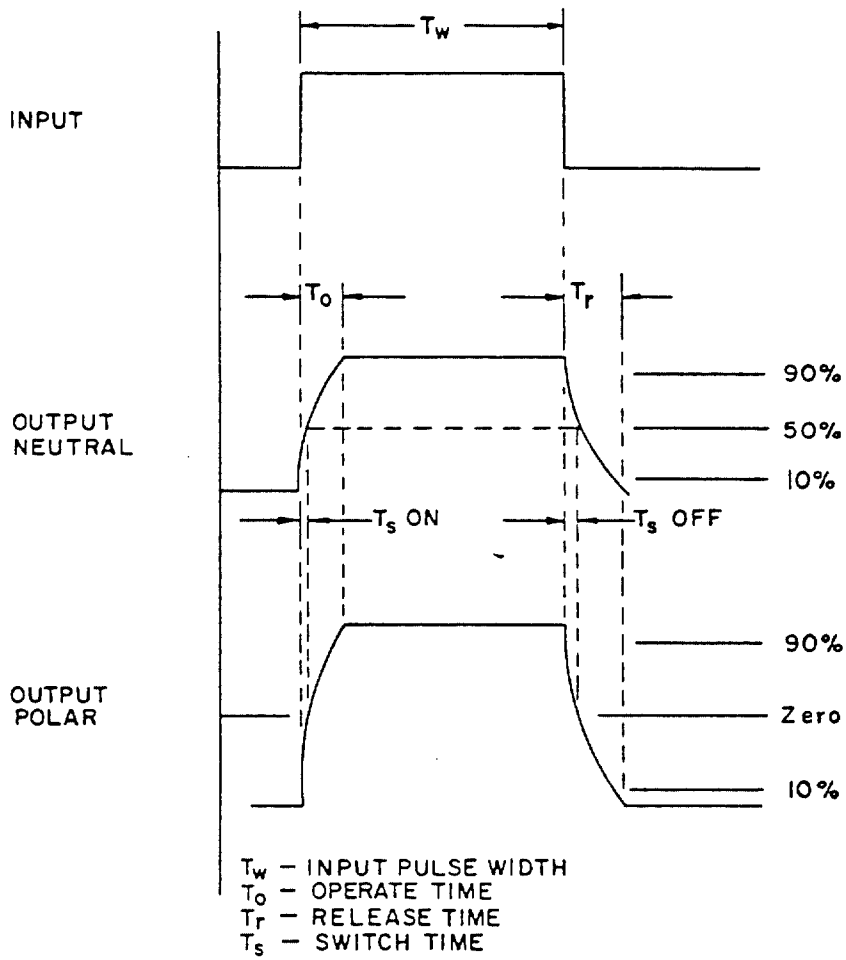


FIGURE 7. Pulse characteristics.

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4.7.2.6 Capacitance (when specified) (see 3.5.7). The capacitance shall be measured using method 305 of MIL-STD-202. The following details apply:

(a) Test frequency - 1 kHz.
(b) Points of measurement A and B on figure 8.

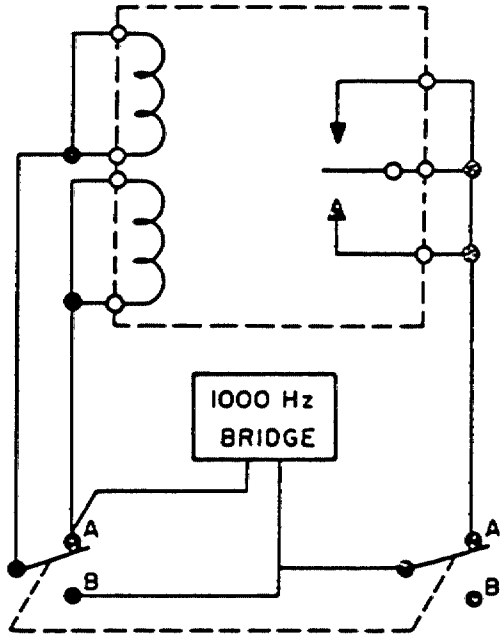


FIGURE 8. Capacitance test circuit.

4.7.3 Thermal shock (see 3.6). The relays shall be tested in accordance with method 107 of MIL-STD-202. The following details apply:

(a) Test condition A.
(b) Measurements of electrical characteristics and dielectric withstanding voltage shall be made.

4.7.4 High and low temperature (see 3.7). The high and low temperature test shall, unless otherwise specified, be applied as follows:

(a) Operating - -10° to +50°C.
(b) Nonoperating and storage - -62° to -71°C.

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4.7.4.1 Method. Prior to application of and at completion of the test, relays shall be tested as specified in 4.7.9, 4.7.10, and 4.7.2.1 through 4.7.2.5 at 25°C. During the test sequence, operation shall be as specified in 4.7.2.5 in the maximum capability mode, that is, universal-relays operating universal, not neutral. The relays shall be in operation (where specified) until satisfactory or unsatisfactory operation is determined. Satisfactory operation shall be a directive to proceed to the next temperature and retest for operation after 1-hour stabilization time. Unless otherwise specified, test sequence is as follows:

Step 1 - - - 25°C operating.
Step 2 - - - -10°C operating.
Step 3 - - - -62°C nonoperating-storage.
Step 4 - - - -10°C operating.
Step 5 - - - 50°C operating.
Step 6 - - - 71°C nonoperating-storage.
Step 7 - - - 25°C operating.

4.7.5 Moisture resistance (see 3.8). Relays shall be tested in accordance with method 106 of MIL-STD-202. The following details and exceptions shall apply:

(a) Mounting - Optional.
(b) Initial measurement - Insulation resistance.
(c) Polarization - 100 volts dc connected between the case ground terminal and all the input terminals of the relay connected together. The case ground terminal shall be negative and the other terminal positive.

NOTE: The polarization requirement shall apply only to one-half the relays being subjected to the test.

(d) Step 7b of MIL-STD-202, method 106 is not required.
(e) Final measurements - Upon completion of step six of the final cycle, and after a 24-hour drying period at a relative humidity of 50 ± 5 percent, measurements of insulation resistance, dielectric withstanding voltage, and electrical characteristics shall be made.

4.7.6 Vibration, high frequency (see 3.9). The relays shall be tested in accordance with method 204 of MIL-STD-202. The following details shall apply:

(a) Relays shall be rigidly mounted in a suitable test jig. The vibration shall be monitored on top of the test jig in the proximity of the relays' support points.
(b) Electrical-load conditions. Unless otherwise specified (see 3.1), the relays shall be vibrated in each of the three mutually perpendicular directions for 1 hour with the input and output circuits deenergized and for 1 hour with the input continuously cycled at 1,200 baud (input and output at maximum specified) (total of 6 hours). Relays shall be monitored for proper operation in the neutral biased condition if possible, or neutral for the last 2 minutes of each cycling period.
(c) Test condition D (except a double amplitude of 0.06 or 10G, whichever is less, shall apply).
(d) Testing after vibration. Relays shall be tested for bias distortion in accordance with 4.7.2.4 and shall be examined for loosening of parts, and other damage.

4.7.7 Shock (specified pulse) (see 3.10). The relays shall be tested in accordance with method 213 of MIL-STD-202. The following details shall apply:

(a) Relays shall be rigidly mounted in a suitable test jig.
(b) Test condition A.
(c) Testing after shock. Relays shall be tested for bias distortion in accordance with 4.7.2.4, and shall be examined for loosening of parts, and other damage.

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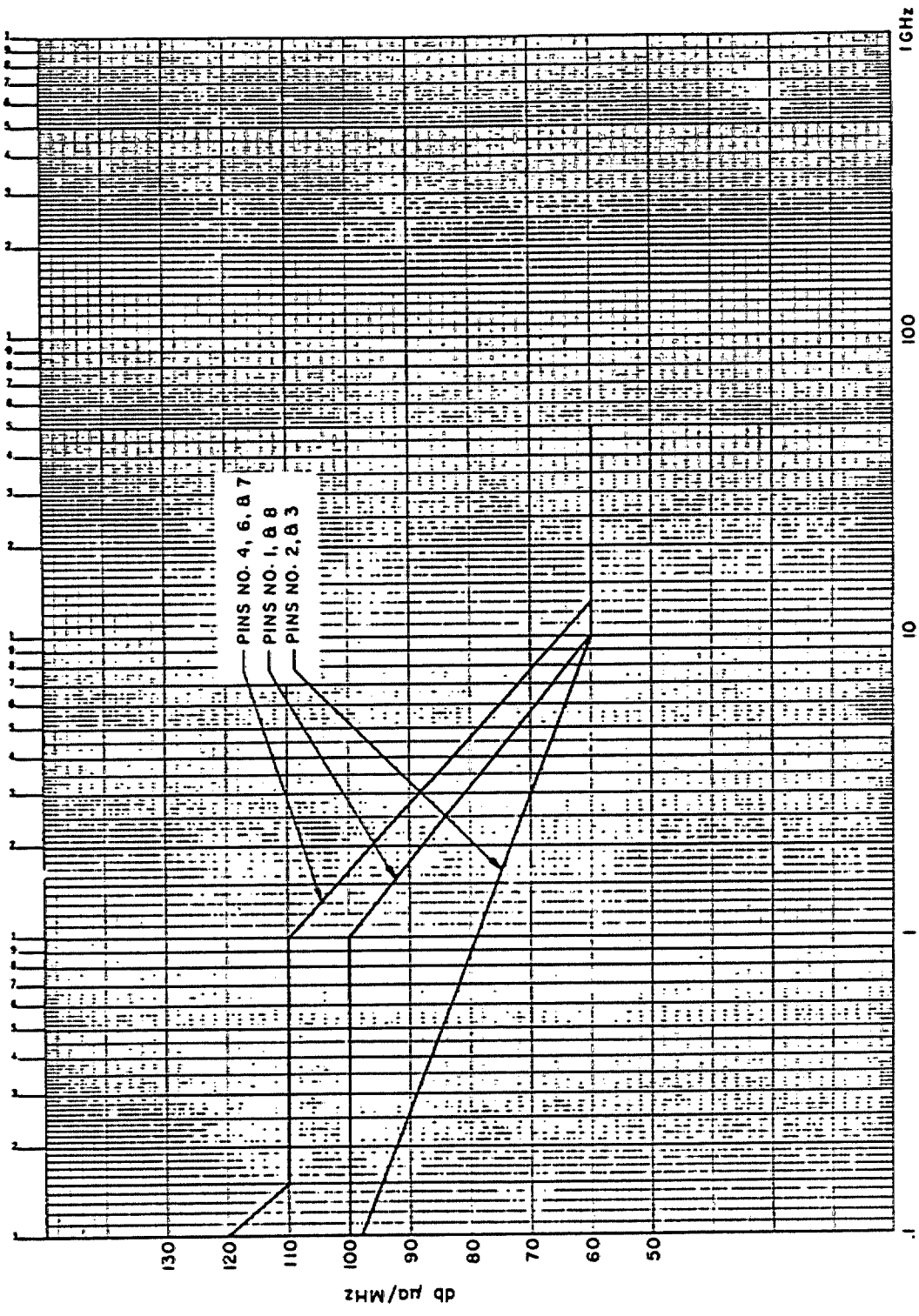


FIGURE 9. Limit for (CE04) conducted broadband emissions for relay pins.

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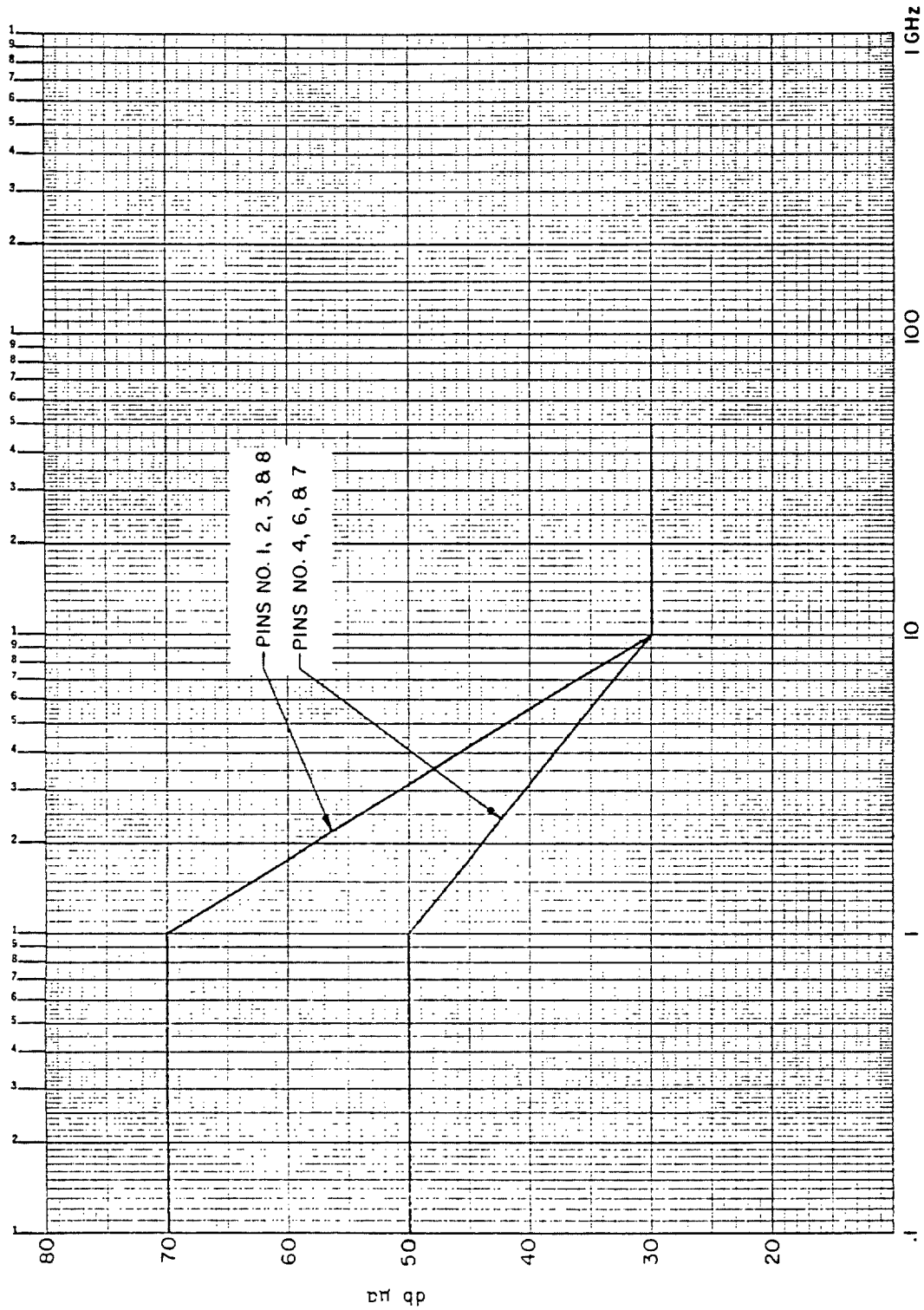


FIGURE 10. Limit for (CE04) conducted narrowband emissions for relay pins.

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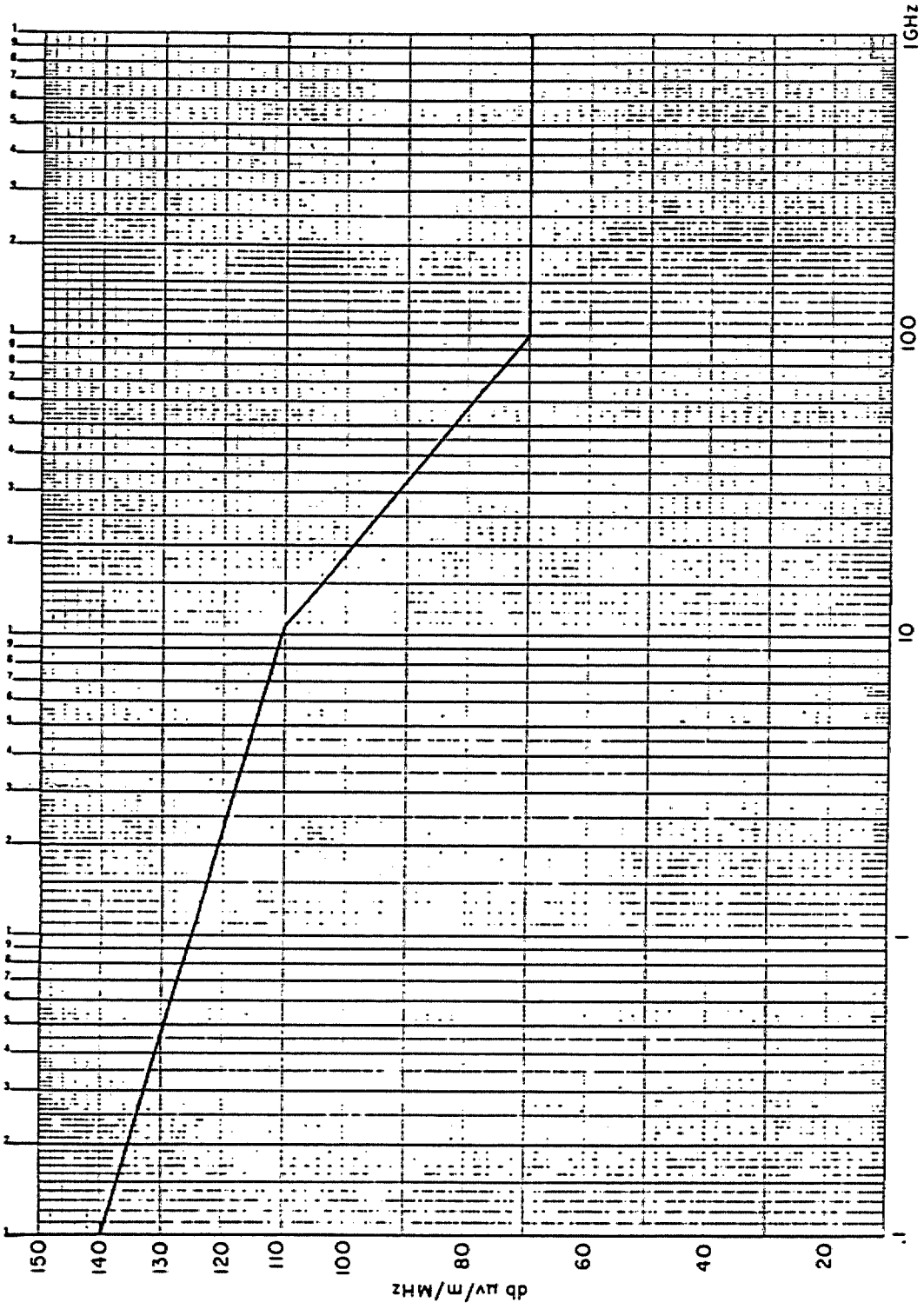


FIGURE 11. Limit for (RE02) radiated broadband emissions.

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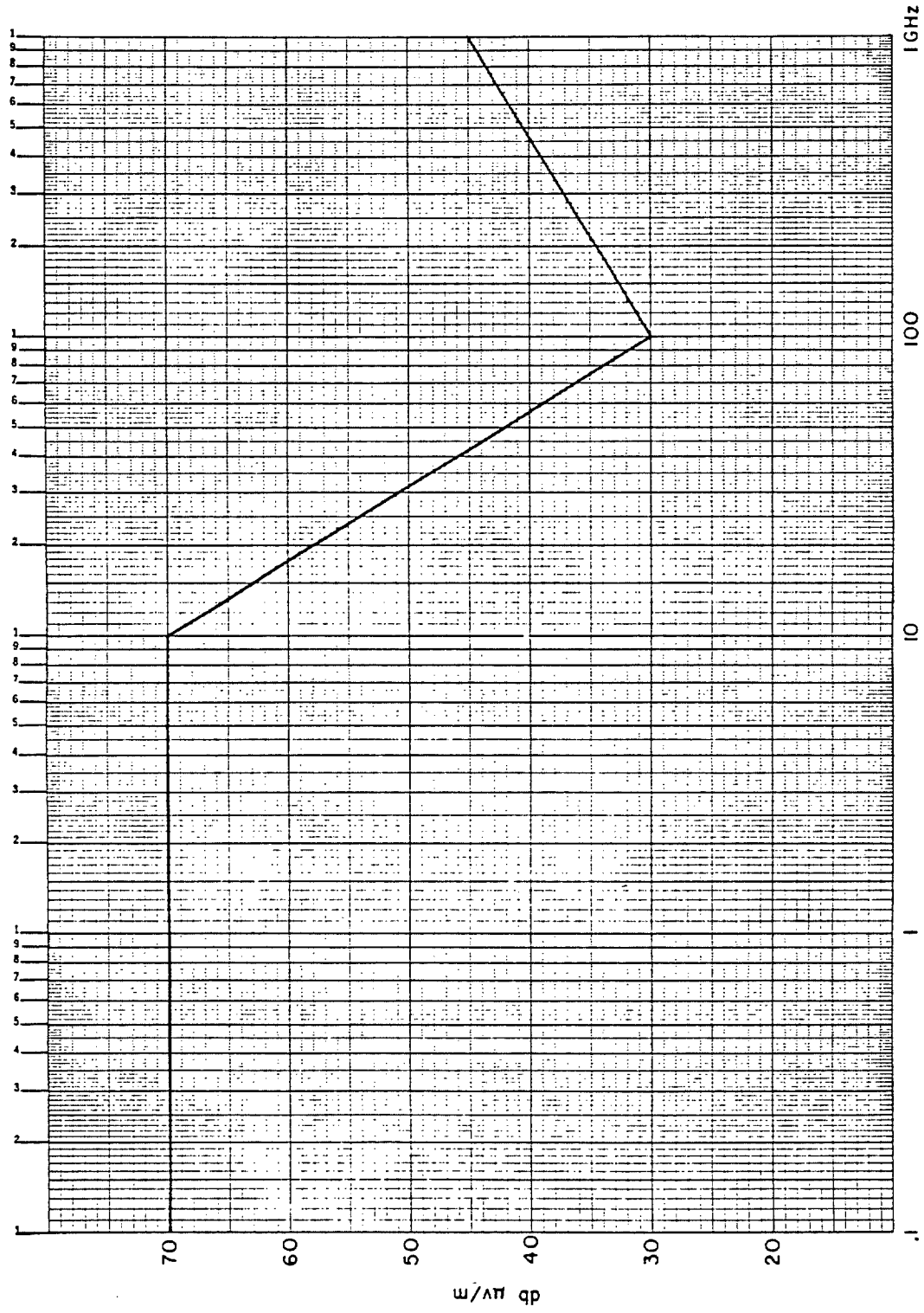


FIGURE 12. Limit for (RE02) radiated narrowband emissions.

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4.7.8 Radio frequency interference test (see 3.11). Relays shall be tested as specified in MIL-STD-461 with the following exceptions:

4.7.8.1 Test method CE04. Limits for broadband emissions shall be as shown in figure 9 in lieu of figure 14 of MIL-STD-461. Limits for narrowband emissions shall be as shown in figure 10 in lieu of figure 15 of MIL-STD-461. The minimum test frequency shall be 100 kHz in lieu of 20 kHz.

4.7.8.2 Test method RE02. Limits for broadband emissions shall be as shown in figure 11 in lieu of figure 22 of MIL-STD-461. Limits for narrowband emissions shall be as shown in figure 12 in lieu of figure 21 of MIL-STD-461.

4.7.8.3 Test method CS06. Limits shall be as specified in figure 19 of MIL-STD-461. All pins of the relay with the exception of pin 5 shall be treated as a power lead and tested accordingly.

4.7.8.4 Test method RS02. The case susceptibility test of RS02 shall be conducted as specified except that the turns around the test sample shall be separated approximately 2 cm. Limits shall be as specified in MIL-STD-461.

4.7.8.5 Test method RS03. The limit shall be as specified in MIL-STD-461, except the maximum frequency shall be 1 GHz in lieu of 10 GHz.

4.7.9 Dielectric withstanding voltage (see 3.12). Relays shall be tested in accordance with method 301 of MIL-STD-202. The test circuit shall be as shown in figure 13. The following details shall apply:

(a) Magnitude and nature of test voltage - 500 ±10 volts rms 60 Hz.
(b) Method of connection of test voltage to specimen - (see figure 13).

After the test, relays shall be examined for any evidence of damage, arcing, and breakdown.

4.7.10 Insulation resistance and isolation (see 3.13). Relays shall be tested in accordance with method 302 of MIL-STD-202. The following details shall apply:

(a) Test condition B.
(b) Electrification time: Until instrument reading is steady or 1 minute, whichever occurs first.
(c) Points of measurement:
    (1) Insulation resistance - As shown in figure 14.
    (2) Isolation - As specified in 3.13.

4.7.11 Terminal strength (see 3.14).

4.7.11.1 Pull. Unless otherwise specified, a force of 5 pounds shall be applied to the terminal at the point where the lead from the external circuit connects to it. The force shall be applied at a right angle to the axis of the terminal. The force shall be increased gradually to the specified magnitude and held at that value for at least 30 seconds. A minimum of two terminals shall be tested. After the test, the relays shall be examined for loosening of terminals and other damage.

4.7.12 Seal (when applicable) (see 3.15). Relays shall be tested in accordance with method 112 of MIL-STD-202. The following details apply:

(a) Test-condition letter - C (procedure III or IV at the option of the manufacturer). Silicone oil shall not be used.
(b) Leakage-rate sensitivity - 10^{-6} atm cc/sec for relays greater than 2 cubic inches in volume and 10^{-8} atm cc/sec for relays 2 cubic inches or less in volume.
(c) Reduced pressure of chamber and duration of pressurization (procedure IV) - Below 5 mm of mercury until leak detector reading is stabilized.
(d) Measurements after test - Not applicable.

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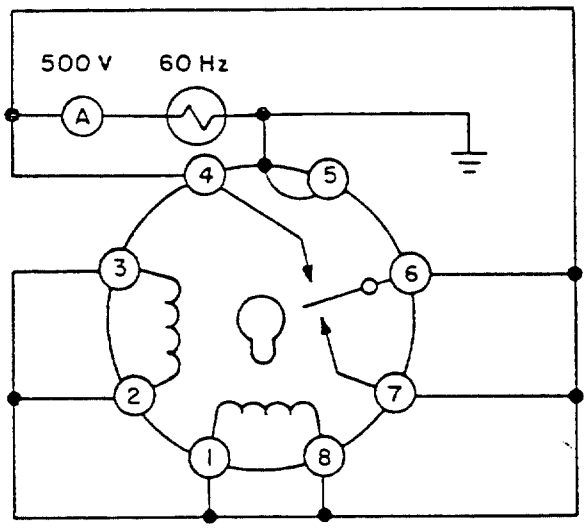


FIGURE 13. Dielectric withstanding voltage.

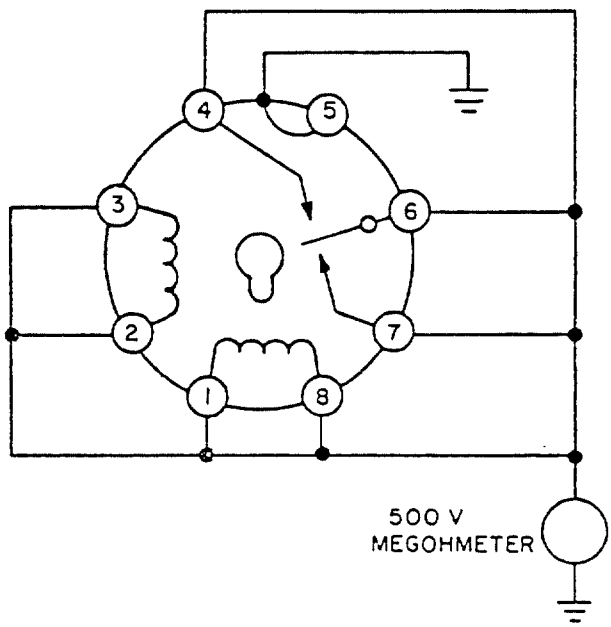


FIGURE 14. Insulation resistance.

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4.7.13 Altitude test (see 3.16). Relays shall be tested for 20 hours in accordance with method 105 of MIL-STD-202; test shall be as in 4.7.14(a)(3). Time spent on this test may be applied as part of required life-test time. Review shall be a part of the sample units subjected to life tests. The following details apply:

(a) Method of mounting - Optional.
(b) Test condition B.

4.7.14 Life (see 3.17). Relays shall be subjected to the following procedure:

(a) (1) With relays mounted in normal mounting means, they shall be driven with the rated rectangular pulse voltage (see 3.1) for 111 hours at 1200 baud. The output shall switch ±130 V polar, and 30 mA current. The duration of the turn-on shall be 50 ±5 percent of each operation of the relay. The ambient temperature of the relay environment shall be maintained at the highest specified operating temperature (see 3.1). The relay shall be monitored for misses (see 6.5.4) during the last hour of each 111-hour step.
(2) Repeat (a)(1) as above for 111 hours, except with the ambient temperature at the lowest specified operating temperature.
(3) Repeat (a)(1) as above for 111 hours, except with the temperature at room ambient.
(b) Same as (a) above, except for each mode of applicable operation, the relay shall be energized continuously with the rated nonoperating current ($^{+0}_{-10}$ percent). The relay shall be monitored to verify that it remains nonoperative.
(c) Same as (a) above, except for each mode of applicable operation, the relay shall be energized continuously with the rated operating current ($^{+10}_{-0}$ percent). The relay shall be monitored to verify that it remains operative.
(d) After the above tests, relays shall be subjected to the following test procedures: Dielectric withstanding voltage, insulation resistance, and all of the electrical tests, in accordance with 4.7.9, 4.7.10, and 4.7.2.1 thru 4.7.2.6, respectively.

5. PREPARATION FOR DELIVERY

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Relays shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. Relays shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservative application. Preservatives shall not be used.

5.1.1.4 Unit packaging. Unless otherwise specified (see 6.2), relays shall be individually packaged in accordance with MIL-P-116, submethod IA-8 insuring compliance with the general requirements paragraph under methods of preservation (unit protection) and the physical protection requirements therein. Each unit package shall be placed in a supplementary container conforming to PPP-B-566 or PPP-B-676.

5.1.1.5 Intermediate packaging. Intermediate packaging is not required.

5.1.2 Level C. Relays shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

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5.2.1 Level A. The packaged relays shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirements. In lieu of the closure and waterproofing requirements in the appendix of PPP-B-636, closure and waterproofing shall be accomplished by sealing all seams, corners and manufacturer's joint with tape, two inches minimum width, conforming to PPP-T-60, class 1 or PPP-T-76. Banding (reinforcement requirements) shall be applied in accordance with the appendix to PPP-B-636 using nonmetallic or tape banding only.

5.2.2 Level B. The packaged relays shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style optional, special requirements. Closures shall be in accordance with the appendix thereto.

5.2.3 Level C. The packaged relays shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

5.2.4.1 Level A. Relays, packed as specified in 5.2.1, shall be unitized on pallets in conformance with MIL-STD-147, load type I, with a fiberboard cap (storage aid 4) positioned over the load.

5.2.4.2 Level B. Relays, packed as specified in 5.2.2, shall be unitized as specified in 5.2.4.1 except that the fiberboard caps shall be class domestic.

5.2.4.3 Level C. Relays, packed as specified in 5.2.3, shall be unitized with pallets and caps of the type, size and kind commonly used for this purpose and shall conform to the applicable carrier rules and regulations.

5.3 Marking. In addition to any special marking required by the contract or purchase order (see 6.2), each unit package, supplementary and exterior container, and unitized load shall be marked in accordance with MIL-STD-129.

5.4 General.

5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2, and 5.2.3) shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Army procurements.

5.4.2.1 Level A unit packaging. MIL-P-116 submethod IC-1 shall be used in lieu of submethod IA-8. All unit and intermediate containers shall either be weather (or water) resistant or overwrapped with waterproof barrier materials. Boxes conforming to MIL-B-43014, which do not require the use of a waterproof overwrap, may be substituted for boxes conforming to PPP-B-566 or PPP-B-676 (see 5.1.1.4).

5.4.2.2 Level A and B packing. For level A packing when quantities per destination are less than a unitized load, the fiberboard containers shall not be banded but shall be placed in a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, class 2, style 4 of PPP-B-585, class 3, style 2 or 3. Closure and strapping shall be in accordance with applicable container specification except that metal strapping shall conform to QQ-S-781, type I, class B. When the gross weight exceeds 200 pounds or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids (laid flat) shall be applied in accordance with the requirements of the container specification. If not described in the container specification, the skids shall be applied in a manner which will adequately support the item and facilitate the use of material handling equipment. For level B packing, fiberboard boxes shall be weather resistant as specified in level A and the containers shall be banded (see 5.2.1 and 5.2.2).

5.4.2.3 Level A and B unitization. For level A and B unitization, the fiberboard caps shall be weather resistant and softwood pallets conforming to NN-P-71, type IV, size 2 shall be used (see 5.2.4.1 and 5.2.4.2).

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5.4.3 Navy procurements. For Navy procurements the use of polystyrene loose fill material (such as strips, strands and beads) is prohibited for packaging and packing applications.

5.4.4 Air Force procurements. For Air Force procurements submethod IC-3 shall be used in lieu of submethod IA-8 (see 5.1.1.4).

6. NOTES

6.1 Intended use. The solid state relays conforming to this specification are intended for use in standard telegraph or data circuits and associated equipment. These relays will be used essentially to replace existing electromechanical relays. (The terms "coil" and "contact" are equivalent to the terms "input" and "output", respectively, for both solid state and electromechanical relays.)

6.2 Ordering data. Procurement documents should specify:

(a) Title, number, and date of this specification.
(b) Title, number, and date of the applicable specification sheet.
(c) Part number (see 3.1).
(d) Hermetically sealed, if applicable (see 3.15).
(e) Initial wiring configuration required (see 3.4.2, 3.4.1.1.1 and 3.4.1.1.2).
(f) Levels of preservation-packaging and packing required (see 5.1 and 5.2).
(g) Method of preservation, if other than submethod IA-8 (see 5.1.1.4).
(h) Special marking, if required (see 5.3).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List (QPL) whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products list is the Department of the Air Force, Rome Air Development Center, Griffiss Air Force Base, Attn: EMTS, NY 13440; however, information pertaining to qualification of products may be obtained from the Defense Electronics Supply Center (DESC-E), Dayton, Ohio 45444. Application for qualification tests shall be made in accordance with provisions governing qualification SD-6 (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

6.4 Intermetallic contact. The finishing of metallic areas to be placed in intimate contact by assembly presents a special problem, since intermetallic contact of dissimilar metals results in electrolytic couples which promote corrosion through galvanic action. To provide the required corrosion protection, intermetallic couples are restricted to those permitted by table IX. Table IX shows metals and alloys (or plates) by groups which have common electromotive forces (EMF) within 0.05 volt when coupled with a saturated calomel electrode in sea-water at room ambient temperatures. All members of a group are considered as completely compatible, one with the other. Compatible couples between groups have been specified in table IX based on a potential difference of 0.25 volt maximum. To simplify any arithmetic involved, table IX shows, in addition to EMF against a calomel electrode, a derived "anodic index" with group 1 (gold, etc.,) as 0 and group 18 (magnesium, etc.,) at 175. Subtraction of a lower group anodic index represents the EMF in hundredths of a volt.

6.4.1 Groups. Table IX sets up 18 primary groups. It may be noted that neither the metallurgical similarity or dissimilarity of metals is the parameter for selection of compatible couples. All members within a group, regardless of metallurgical similarity, are considered inherently nonsusceptible to galvanic action, when coupled with any member within the group; for example, such dissimilar metals as platinum and gold. Similarly, such basically dissimilar alloys as austenitic stainless steel, silver-solder, and low brass (all members of group 5) are inherently nonsusceptible when coupled together.

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TABLE IX. Compatible couples (see 6.4) 1/

Group No.	Metallurgical category	EMF (volt)	Anodic index (0.01 v)	Compatible couples
1	Gold, solid and plated; gold-platinum alloys; wrought platinum (most cathodic)	+ 0.15	0	
2	Rhodium plated on silver-plated copper	+ 0.05	10	
3	Silver, solid or plated; high silver alloys	0	15	
4	Nickel, solid or plated; monel metal, high nickel-copper alloys	- 0.15	30	
5	Copper, solid or plated; low brasses or bronzes; silver solder; German silver; high copper-nickel alloys; nickel-chromium alloys; austenitic corrosion-resistant steels	- 0.20	35	
6	Commercial yellow brasses and bronzes	- 0.25	40	
7	High brasses and bronzes; naval brass; Muntz metal	- 0.30	45	
8	18 percent chromium type corrosion-resistant steels	- 0.35	50	
9	Chromium, plated; tin, plated; 12 percent chromium type corrosion-resistant steels	- 0.45	60	
10	Tin-plate; terneplate; tin-lead solder	- 0.50	65	
11	Lead, solid or plated; high lead alloys	- 0.55	70	
12	Aluminum, wrought alloys of the duralumin type	- 0.60	75	
13	Iron, wrought, gray, or malleable; plain carbon and low alloy steels, armco iron	- 0.70	85	
14	Aluminum, wrought alloys other than duralumin type; aluminum, case alloys of the silicon type	- 0.75	90	
15	Aluminum, cast alloys other than silicon type; cadmium, plated and chromated	- 0.80	95	
16	Hot-dip-zinc plate; galvanized steel	- 1.05	120	
17	Zinc, wrought; zinc-base die-casting alloys; zinc, plated	- 1.10	125	
18	Magnesium and magnesium-base alloys, cast or wrought (most anodic)	- 1.60	175	

1/ Compatible couples - potential difference of 0.25 volt maximum between groups.

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6.4.2 Compatibility graphs. Permissible couple series are shown in table IX by the graphs at the right. Members of groups connected by lines will form permissible couples. The "●" indicates the most cathodic member of each series, the "○" an anodic member; and the arrow indicates the anodic direction.

6.4.3 Selection of compatible couples. Proper selection of metals in the design of equipment will result in fewer intermetallic contact problems. For example, for sheltered exposure, neither silver nor tin require protective finishes. However, since silver has an anodic index of 15 and tin 65, the EMF generated as a couple is 0.50 volt, which is not allowable. In this case, other metals or plates will be required. It should be noted that, in intermetallic couples, the member with the higher anodic index is anodic to the member with the lower anodic index and will be susceptible to corrosion in the presence of an electrolytic medium. If the surface area of the cathodic part is significantly greater than that of the anodic part, the corrosive attack on the contact area of the anodic part may be greatly intensified. Material selection for intermetallic contact parts, therefore, should establish the smaller part as the cathodic member of the couple, whenever practicable.

6.4.4 Plating. When base metals intended for intermetallic contact form couples not allowed by table IX, they are to be plated with those metals which will reduce the potential difference to that allowed by table IX.

6.5 Definitions.

6.5.1 Baud (the unit of modulation rate). One baud corresponds to a rate of one unit interval per second. The modulation rate is expressed as the reciprocal of the duration in seconds of the unit interval. Example: if the duration of the unit interval is 20 milliseconds, the modulation rate is 50 bauds.

6.5.2 Hermetic seal. A hermetic seal is metal-to-metal, ceramic-to-metal or glass-to-metal only.

6.5.3 Mark, or mark time. In telegraphy, equivalent to "signal", "energized" or "actuated" condition. Opposite of "space" (also see 6.5.12).

6.5.4 Miss. A miss is a failure to respond to the input test signal within the allowable distortion limits and on the specified operation limits of the relay.

6.5.5 Module, bias. The bias module is a component part of a universal relay (neutral, bias input, polar/neutral output) which provides the sampling characteristic for neutral to polar operation. The bias module provides input control for two receiving modules, thereby establishing a predetermined switching level which is a function of the external dc bias current setting.

6.5.6 Module, receiving. The receiving module is a component part of a solid state relay that provides the input characteristics of a specified relay (neutral, polar, or neutral/polar), and provides the drive power to the transmitting module.

6.5.7 Module, transmitting. The transmitting module is a component part of a solid state relay that receives its input drive from a receiving module and presents a closed or open-contact characteristic to a telegraph line. The contact configuration of the transmitting module may be either neutral or polar.

6.5.8 Pulse-width distortion. Relative input-output pulse width difference expressed as a percentage of the input pulse width. Measurements shall be at 50-percent amplitude levels for neutral pulses and at zero amplitude levels for polar signals.

$$\text{Example: Percent distortion} = 100 \times \frac{\text{(Switch time OFF - switch time ON)}}{\text{Input pulse width}}$$

6.5.9 Relay, neutral. A general purpose relay requiring one actuating signal which controls one-form A contact configuration (single-pole single-throw normally-open).

6.5.10 Relay, polar. A polarized, two-input, null-seeking relay with form-K contact configuration (single-pole double-throw, normally-open).

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6.5.11 Relay, universal. A two-input relay of contact-K output configuration (single-pole double-throw center-off) which can be operated in the following modes:

(a) Output may be polar or neutral.
(b) Input may be polar or neutral.
(c) Signal may be applied to either relay input.

6.5.12 Sensitivity (input current). The differential in input current required to assure that the output transfers from operating to nonoperating condition and vice versa. As a design objective, the values from DCA Circular 330-175-1 (Engineering-Installation Standards Manual) are as follows:

Input mode	Input current	
	Output (operating)	Output (nonoperating)
Neutral	$\geqq$ 32 mA	$\leqq$ 28 mA
Polar	$\geqq$ +2 mA	$\leqq$ -2 mA

6.5.13 Space, or space-time. Equivalent to "no signal", "unenergized", or "released" condition. Opposite of "mark".

6.5.14 Switch-time. Switch-time is defined as the time-lapse between a step function input signal and a specified contact current amplitude, expressed as a percentage of the test current condition(s). Switch-time applies to leading and trailing edge measurements. Resistive input and output circuit conditions shall apply.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - EL
Navy - EC
Air Force - 17

Review activities:
Army - MU, EL, MI
Navy - SH
Air Force - 11, 13, 80, 89
DSA - ES

User activities:
Army - SM
Navy - MC, MO, AS, OS
Air Force - 14, 19

Preparing activity:
Air Force - 17

Agent:
DSA - ES

(Project 5945-0288)

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